TWELVE PAIRWISE ADJACENT FACES: A CHIROTOPE REDUCTION TO BOOLEAN UNSATISFIABILITY

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Abstract. Can twelve planar faces form a closed surface in three-dimensional space with every pair sharing an edge? We turn this geometric existence question into explicit finite proof obligations and provide a workflow for checking a distributed computation. We require an embedded surface in $\mathbb{R}^3$. Faces are simple planar disks with distinct supporting planes, and any two faces intersect in precisely one common edge. Such a surface is trivalent and has genus 6. We reduce its existence to the satisfiability of one of 59 explicit Boolean formulas, one for each orientable neighbourly map on twelve faces.

The reduction first puts a realization into general position without changing its combinatorics. The twelve supporting planes and the plane at infinity then define a uniform rank-4 chirotope on 13 elements. Bracket signs encode necessary conditions for face simplicity, boundary-crossing parity and local vertex geometry. A reorientation gauge removes redundant signs. Each resulting formula has 5 257 variables and 52 177 clauses.

We give the necessity proofs, recompute the 59 combinatorial types, and describe exact geometric controls and a separate reference encoder. We prove that refuting every leaf of a finite tree of case splits rooted at the unrestricted formula refutes the formula. We also specify a protocol for producing LRAT refutations of such leaves, checking them with a formally verified checker, and verifying the computation independently by rechecking or recomputing its leaves. Unsatisfiability of all 59 formulas would exclude a surface in the stated model. A complete set of refutations has not been obtained, so this non-existence conclusion remains conditional. Conversely, a satisfying assignment would not by itself establish geometric realizability.

1. Introduction

1.1. **The question.** Call a closed surface in $\mathbb{R}^3$ that is a finite union of planar polygons a *polyhedral surface*, and call it *neighbourly* when every two of its faces share an edge. The tetrahedron is neighbourly with four faces. Szilassi's toroidal polyhedron [21] is neighbourly with seven. These are the only examples reported in [2]. We study the next combinatorially admissible case within the model specified below:

> **Question.** Is there a closed embedded polyhedral surface in $\mathbb{R}^3$ satisfying (H1)–(H3) below, with twelve faces, every two of which share an edge?

The restriction to twelve is not arbitrary. Pairwise adjacency forces $E = \binom{n}{2}$ edges and, as we show in Lemma 2.4, every vertex of such a surface has degree three, whence $V = n(n-1)/3$ and

$$g \;=\; \frac{2-\chi}{2} \;=\; \frac{(n-3)(n-4)}{12}.$$

A closed surface embedded in $\mathbb{R}^3$ is orientable, so g must be a non-negative integer, which happens exactly for $n \equiv 0, 3, 4, 7 \pmod{12}$. After 4 and 7 the next admissible value is $n = 12$, with $V = 44$, $E = 66$ and genus 6. That the corresponding *combinatorial*

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objects exist in abundance is classical: by the Ringel–Youngs solution of the Heawood problem [17, 16], the complete graph K_n embeds in the orientable surface of genus $\lceil (n-3)(n-4)/12 \rceil$, and when the unrounded quotient $(n-3)(n-4)/12$ is an integer — exactly for these n — the embedding is a triangulation. The question is whether one of those combinatorial objects can be realized by flat faces in space without self-intersection.

1.2. **What is known.** Dualizing a neighbourly polyhedron with 12 faces gives a triangulation of the orientable surface of genus 6 with 12 vertices in which every two vertices are joined by an edge; 12 is the minimum number of vertices for that surface: a triangulation with v vertices has $3(v+10)$ edges, at most $\binom{v}{2}$, which forces $v \geq 12$ (see [12] for the general minimum-vertex theorem). Altshuler, Bokowski and Schuchert [1] classified these: there are exactly 59 combinatorial types, all orientable of genus 6. They also proved several properties that a polyhedral realization would have to have, and left realizability open. Schewe [19] settled the realizability of the dual objects negatively: no triangulation of the genus-6 surface with 12 vertices has a polyhedral realization in $\mathbb{R}^3$. His Theorem 1.1 follows from Proposition 1.2: none of the 59 types admits an acyclic uniform rank-4 oriented matroid *admissible for that triangulation*. Admissibility excludes circuits witnessing an intersection of a triangle with a disjoint edge ([19, Section 2]); a SAT solver discharges this constrained combinatorial problem.

Schewe's theorem does not answer our Question. Polyhedral realizability is not preserved by combinatorial duality: his objects are 12 *points* carrying 44 triangular faces, ours are 12 *planes* carrying 44 trivalent vertices, and a realization of either does not provide a realization of the other by combinatorial duality alone. The general question — which complete graphs occur as face-adjacency graphs of closed polyhedral surfaces — is recorded as open in [2], where the tetrahedron and the Szilassi polyhedron are named as the only known instances.

1.3. **Contribution.** This paper gives an explicit reduction of the Question to a finite Boolean computation. The geometric necessity arguments are proved below; the topological and oriented-matroid facts used in them are identified by references. Specifically:

(1) **Case list.** Every neighbourly polyhedron with twelve faces realizes one of exactly 59 neighbourly maps (Theorem 3.8). We recompute the list by an exhaustive search over oriented triple systems; its size agrees with [1], and its classes are, one by one, those of the published list [14].
(2) **General position.** Every realization may be perturbed into one in which no four of the thirteen covectors (twelve face planes and the plane at infinity) are linearly dependent (Theorem 4.2). The signs of the $\binom{13}{4} = 715$ brackets are then defined and form a uniform rank-4 chirotope.
(3) **A sign dictionary.** Every geometric predicate the argument needs — the order of the points marked on a line, betweenness inside an edge, the turn at a corner, the parity of boundary crossings along a ray — is an explicit Boolean combination of bracket signs (Section 5). The identities are exact, not approximations.
(4) **Five layers of necessary conditions.** We define a CNF formula Φ_T for each map T as the conjunction of five layers S, H, A, B, I and a gauge, and prove that every general-position realization of T, after the gauge normalization of Corollary 8.2, satisfies all of them (Theorem 9.2). Layers S, H and A are exactly the statement that the bracket signs form a uniform rank-4 chirotope, encoded through 78 linear orders rather than through 411 840 six-literal clauses; layers B and I are geometric: the boundary polygon of each face is simple, the traversal orientation of each face is the one its crossing parity forces, and no vertex carries two reflex corners.
(5) **Conditional non-existence.** If all 59 formulas are unsatisfiable, the Question has a negative answer (Corollary 9.3).

(6) **Coverage by cube trees.** A formula is unsatisfiable as soon as every leaf of a finite tree of case splits rooted at the empty cube is refuted (Theorem 9.9); a refuted tree rooted at a nonempty cube settles only that region (Remark 9.12).
(7) **Verification protocol.** We describe the discipline under which the refutations are to be produced and checked, so that the eventual computational claim rests on a formally verified proof checker [22] rather than on a solver's word, and so that a third party can verify it by rechecking retained proofs or, without access to stored proofs, by recomputing any or all of its leaves (Section 10).

1.4. **What is not claimed.** The Boolean computation has *not* been carried out. Theorem 9.2 and Corollary 9.3 are a reduction and a conditional statement; the Question remains open. Two further limitations are worth stating at the outset. First, only necessity is encoded: a satisfying assignment of some Φ_T would not produce a polyhedron, because the layers are necessary conditions and because a uniform chirotope need not be realizable over $\mathbb{R}$. Second, the reduction is stated under the hypotheses (H1)–(H3) of Section 2, which are the regular polygonal-cell conventions used throughout this paper; dropping them — allowing faces with holes, two faces to meet in more than one edge, or additional contacts between faces — changes the object under study, and those variants are not settled by these 59 formulas.

A numerical search for the same twelve-face objects is described by Bokowski and Kevin H. (CodeParade) [6, Sections 4 and 8.3]. It found no embedded realization among the 59 dual types and presents an intersecting example in Figure 38; this is a search result, not a non-existence proof. The associated NeighborlyPolyhedra repository [10] documents the search and further near-misses. Our approach instead gives finite necessary conditions whose certified unsatisfiability would imply non-existence. The cited 2025 paper leaves the twelve-face question unresolved.

1.5. **Organization.** Section 2 fixes the hypotheses and proves the local facts: trivalence, the Euler count, and the vertex lemma that at most one of the three corners at a vertex is reflex. Section 3 defines neighbourly maps and realizations and proves the case list. Section 4 proves the general-position theorem. Section 5 develops the sign dictionary. Sections 6 and 7 define and justify the five layers, Section 8 the gauge. Section 9 assembles the formula, states the reduction and proves that refuted trees of cases cover the formula, and Section 10 describes the verification protocol. Section 11 states precisely what remains to be done. Appendix A contains the numerical data, and Appendix B describes the accompanying software.

Acknowledgement of tooling. The accompanying implementation is written in Python with no runtime dependencies beyond the standard library; it uses CaDiCaL [4] as a proposing solver and `cake_lpr` [22] as the accepting checker. The partition heuristic follows [3].

2. Polyhedra, hypotheses and local structure

Definition 2.1. A *polyhedron* is a compact closed surface $\Sigma \subset \mathbb{R}^3$ without self-intersections that is a finite union of closed planar polygons, its *faces*. Each face is a closed simple polygonal disk and is the closure in Σ of a maximal connected relatively open subset of Σ contained in an affine plane. *Edges* are the maximal straight segments of the union of face boundaries; degree-two subdivision points are suppressed, so faces and edges form a finite regular cell decomposition of Σ. Two faces *share an edge* when their intersection is an edge of both. The polyhedron is *neighbourly* when every two faces share an edge.

Three hypotheses are used throughout. Each is a standard convention, and each marks a place where a wider reading of the word *polyhedron* would leave a case uncovered.

Hypothesis 2.2.

(H1) The intersection of two faces is a common face of both: empty, a single common vertex, or a single common edge. This is our polygonal-cell intersection convention; it forces *exactly* one shared edge per pair and, with Lemma 2.4, trivalent vertices.
(H2) Each face is a simple polygon: one boundary cycle, no holes.
(H3) Distinct faces have distinct supporting planes.

Remark 2.3. Within the closed embedded polygonal-cell model of Definition 2.1, maximality of faces is equivalent to requiring that no two edge-adjacent faces be coplanar. Indeed, two coplanar faces sharing an edge would have interiors joined through an interior point of that edge inside a connected relatively open planar subset of Σ, contradicting maximality. Conversely, if adjacent faces are noncoplanar, no relatively open planar subset of Σ can contain an edge-interior point: its neighbourhood contains open portions of both incident faces. Nor can such a subset contain a vertex, since any neighbourhood of the vertex contains edge-interior points. Thus the connected relatively open planar patches are contained in individual face interiors, whose closures are the faces. For a neighbourly surface every pair is edge-adjacent, so this condition is precisely (H3).

The model in [2, Section 1, pp. 1430–1431] is broader: its polygonal complexes may have boundary and may have coplanar cells (indeed, Proposition 3.2 constructs flat complexes). It imposes neither our maximality convention nor (H3). The equivalence just proved applies after restricting to a closed embedded surface with our cell conventions and excluding coplanar adjacent cells; it is not an equivalence with that broader model. We cite its Section 2, p. 1435, for the open question about closed surfaces.

Lemma 2.4 (Trivalence)**.** *In a neighbourly polyhedron satisfying* (H1)*, every vertex has degree three.*

Proof. Let v be a vertex. Because Σ is a surface with a regular cell decomposition, the faces incident to v occur in a cyclic order around v, consecutive ones sharing an edge through v; a degree-two vertex would make its two incident faces share both incident edges, contrary to (H1). Thus the degree d of v is at least three. Suppose $d \geq 4$ and let F^i, F^j be two incident faces that are not consecutive in that cyclic order. Their closed sectors at v meet only in v, so no edge through v is common to both. By neighbourliness they share an edge E, necessarily not incident to v. Then $F^i \cap F^j$ contains the edge E and also the point $v \notin E$. By (H1) it is a single common face of both — empty, a vertex or an edge — and, containing E, it would have to be an edge containing E, that is E itself, since distinct edges meet at most in endpoints. But E does not contain v, contradicting (H1). □

Proposition 2.5 (Euler count)**.** *Let Σ be a neighbourly polyhedron with $n \geq 4$ faces satisfying* (H1)*. Then*

$$E = \binom{n}{2}, \qquad V = \frac{n(n-1)}{3}, \qquad \chi = n - \frac{n(n-1)}{6}, \qquad g = \frac{(n-3)(n-4)}{12},$$

and g is a non-negative integer, so $n \equiv 0, 3, 4, 7 \pmod{12}$. For $n = 12$: $V = 44$, $E = 66$, $\chi = -10$ and $g = 6$.

Proof. Each pair of faces shares exactly one edge and each edge lies in exactly two faces, so $E = \binom{n}{2}$. By Lemma 2.4 every vertex has degree three, and counting vertex–edge incidences gives $3V = 2E$, whence $V = n(n-1)/3$. Then $\chi = V - E + F = n - n(n-1)/6$ and $2 - \chi = (n-3)(n-4)/6$. A compact closed surface embedded in $\mathbb{R}^3$ is orientable — a closed non-orientable surface has torsion in $H_1(-;\mathbb{Z})$, which Alexander duality forbids for a subspace of $\mathbb{R}^3$; see [11, §3.3, Corollary 3.46 and the remarks following it] — and it is connected, because any two faces share an edge; so $\chi = 2 - 2g$ with $g \in \mathbb{Z}_{\geq 0}$, and

$g = (n-3)(n-4)/12$. Finally $12 \mid (n-3)(n-4)$ holds exactly for $n \equiv 0, 3, 4, 7 \pmod{12}$, as one checks on the twelve residues. □

The last local fact we need is about reflex corners. Let v be a vertex of a neighbourly polyhedron, incident to the three faces F^f, F^g, F^h. Near v each of them is a plane wedge, and the corner of F^f at v is *reflex* when its interior angle exceeds π.

Lemma 2.6 (At most one reflex corner)**.** *At every vertex of degree three of a polyhedron satisfying* (H1)–(H3)*, at most one of the three corners is reflex.*

Proof. Let v be incident to F^f, F^g, F^h and suppose the corners of F^f and F^g at v are both reflex. Both faces contain the edge $E(f,g)$, which lies on the line $L = h_f \cap h_g$ and has v as an endpoint; let R be the ray of L issuing from v opposite to $E(f,g)$. Note first that $F^f \cap F^g = E(f,g)$: by (H1) that intersection is a common face of both — empty, a vertex or an edge — and it contains the edge $E(f,g)$.

Near v the face F^f is the wedge bounded by the germs of $E(f,g)$ and $E(f,h)$. A plane wedge has interior angle exceeding π if and only if its interior contains the ray opposite to each of its two bounding rays; in particular a reflex corner of F^f at v puts a germ of R in the interior of F^f. The same applies to F^g. Hence $F^f \cap F^g$ contains points of R other than v, while $E(f,g)$ meets R only in v; a contradiction. □

Remark 2.7. The software additionally checks the local conclusion by enumerating the 256 unions of the eight octants cut out by three independent planes. The admissible sector-size profiles are $(1,1,1)$ and $(1,1,3)$, occurring 16 and 48 times respectively; an entry 3 means a reflex corner. The enumeration and its incidence conventions are documented with the code. It is a supplementary check; the reduction uses the proof above.

3. Maps, realizations and the case list

3.1. Neighbourly maps.

Definition 3.1. A *neighbourly map* on $n \geq 4$ faces is a set T of 3-element subsets of $\{1, \dots, n\}$, called *vertices*, such that

- (M1) every pair $\{i,j\}$ lies in exactly two members of T, written $\mathrm{Ends}(i,j) = \{k,l\}$ where $\{i,j,k\}, \{i,j,l\} \in T$; and
- (M2) for every f the *link* $\mathrm{lk}(f)$ — the graph on $\{1,\dots,n\}\setminus\{f\}$ with $a \sim b$ iff $\{f,a,b\} \in T$ — is a single cycle of length $n-1$.

The map is *orientable* when the $|T| = n(n-1)/3$ triples can be cyclically oriented so that every ordered pair (a,b) with $a \neq b$ occurs in exactly one oriented triple.

Condition (M1) says $|T| = n(n-1)/3$ and makes T the triangle set of a two-dimensional simplicial complex on the labels $1, \dots, n$ in which every two labels span an edge and every edge lies in exactly two triangles. Each link $\mathrm{lk}(f)$ is then 2-regular, a disjoint union of cycles, and a label whose link has more than one cycle would be a singular point of the complex. Condition (M2) excludes such points: it makes the complex a triangulation of a closed surface. The following lemma is what lets us pass from geometry to T.

Lemma 3.2. *Let Σ be a neighbourly polyhedron with n faces satisfying* (H1)–(H2)*, its faces labelled $1, \dots, n$. Then the set T of triples of faces meeting at a vertex is a neighbourly map on n faces, and distinct vertices give distinct triples.*

Proof. By Lemma 2.4 each vertex is incident to exactly three faces, so each vertex gives a 3-subset. Each pair $\{i,j\}$ shares exactly one edge, and that edge has exactly two endpoints, so $\{i,j\}$ lies in exactly two triples counted with multiplicity: (M1) holds once we know the triples of distinct vertices are distinct. Suppose $v \neq w$ were vertices with the same incident faces $\{f,g,h\}$. Then v and w are both endpoints of the edge $E(f,g)$, and

likewise of $E(f,h)$ and of $E(g,h)$. Three distinct straight segments cannot have the same pair of endpoints, so two of these edges coincide, contradicting the regularity of the cell decomposition. Finally (M2) is the statement that the faces around a fixed face f close up into one cycle, which holds because the boundary of the disk F^f is a single closed polygon whose edges are $E(f,u)$, $u \neq f$, each consecutive pair meeting at a vertex. □

3.2. **Realizations.** Write a plane as its covector: $h = \{x \in \mathbb{R}^3 : q \cdot (x,1) = 0\}$ with $q = (a,b,c,d) \in \mathbb{R}^4$, $(a,b,c) \neq 0$; its *normal* is $n(q) = (a,b,c)$. Adjoin the *plane at infinity* $q_0 = (0,0,0,1)$, which no affine point satisfies. For a 4-subset $S \subseteq \{0,1,\dots,n\}$ let

$$[S] \;=\; \det\big(q_s\big)_{s\in S}$$

with the rows taken in increasing order of the labels; for an ordered tuple the bracket changes sign with the parity of the permutation. Write $P_{ijk} = h_i \cap h_j \cap h_k$ when that intersection is a single point and $L_{ij} = h_i \cap h_j$ when that intersection is a line.

Definition 3.3. A *realization* of a neighbourly map T on n faces is a tuple $q = (q_1,\dots,q_n)$ of covectors such that, writing $E(i,j)$ for the segment of L_{ij} between P_{ijk} and P_{ijl} with $\{k,l\} = \mathrm{Ends}(i,j)$, and F^f for the closed region bounded in h_f by the closed polygon with vertex sequence $\big(P_{f,u,\,\mathrm{lk}\text{-successor of }u}\big)_{u\in\mathrm{lk}(f)}$:

- (R1) for every $\{i,j,k\} \in T$ the normals $n(q_i), n(q_j), n(q_k)$ are linearly independent, so P_{ijk} exists, and the $n(n-1)/3$ points so obtained are pairwise distinct;
- (R2) for every f the polygon above is simple, so F^f is a closed disk;
- (R3) $F^i \cap F^j = E(i,j)$ for all $i \neq j$; and
- (R4) the planes $h_1,\dots,h_n$ are pairwise distinct.

Lemma 3.4. *The union $\Sigma = \bigcup_f F^f$ of the faces of a realization of a neighbourly map T is a neighbourly polyhedron with n faces satisfying* (H1)–(H3)*, whose induced map is T. Conversely every such polyhedron, with its faces labelled, arises this way from the tuple of its supporting covectors.*

Proof. Let q be a realization. Each F^f is a closed disk by (R2), and two of them meet exactly in the common edge $E(i,j)$ by (R3), which is an edge of both by construction. No fourth face F^i contains a vertex P_{fgh}: by (R3) this point would lie on the three distinct boundary edges $E(i,f), E(i,g), E(i,h)$ of the simple polygon F^i, whereas a point of a simple polygon belongs to at most two of its edges. At P_{fgh} the three incident edge lines are distinct by (R1), so no two incident segments join into one straight edge; consequently the edges are exactly the segments $E(i,j)$. Thus (H1) holds and the faces are pairwise adjacent. Nor can a third face F^k meet the relative interior of $E(i,j)$: by (R3) the point would lie on both $E(k,i)$ and $E(k,j)$, which by (R2) can meet only at their common polygon vertex P_{ijk}, an endpoint of $E(i,j)$. (H2) is (R2) and (H3) is (R4). Σ is compact and closed as a finite union of disks whose boundary edges are each shared by exactly two of them. It is a 2-manifold: an interior point of a face has a disk neighbourhood; a point in the relative interior of $E(i,j)$ has a neighbourhood consisting of two half-disks glued along a segment; and at a vertex P_{fgh} the three wedges of F^f, F^g, F^h meet pairwise in the germs of $E(f,g), E(f,h), E(g,h)$, so the link of the vertex in Σ is a triangle, that is, a circle. The relative interior of each F^f is a connected relatively open planar subset of Σ. Every relatively open neighbourhood in Σ of a point of ∂F^f contains two-dimensional pieces of two incident faces with distinct supporting planes, by (R4), and therefore is not planar. Thus this interior is maximal among connected relatively open planar subsets of Σ, and its closure is F^f. Conversely, a neighbourly polyhedron with labelled faces gives covectors q_i of its supporting planes, and Lemma 3.2 gives the map. At an incident triple, the three normals are linearly independent. Indeed, otherwise their rank is two: the planes are distinct and pass through the same vertex. Their pairwise intersection lines would then

coincide. The three distinct incident edge germs would lie on the two rays of that line, forcing two germs to overlap, contrary to the regular embedded cell structure. Thus each incident triple meets at a unique point, and the vertex of Σ with incident faces $\{i, j, k\}$ is P_{ijk}; by Lemma 3.2 these $n(n-1)/3$ vertices are distinct, which is (R1). The common edge of F^i and F^j lies on L_{ij} and its endpoints are the two vertices whose triples contain $\{i, j\}$, so it is the segment $E(i, j)$ of Definition 3.3. As in the proof of Lemma 3.2, the boundary of the face F^f consists of the edges $E(f, u)$, $u \neq f$, consecutive ones meeting at the vertices $P_{f,u,\mathrm{succ}(u)}$ in the order of $\mathrm{lk}(f)$; it is therefore the polygon of Definition 3.3, which is simple by (H2), and the face, a closed disk in h_f, is the region that polygon bounds. This is (R2), and the regions F^f of the definition are the faces of Σ. By (H1) the intersection $F^i \cap F^j$ is a single common face of both; it contains the edge $E(i, j)$, and distinct edges meet at most in endpoints, so $F^i \cap F^j = E(i, j)$, which is (R3). Finally (R4) is (H3). □

3.3. **The case list.**

Lemma 3.5 (Oriented form)**.** *Let $\vec{T}$ be a set of $n(n-1)/3$ cyclically ordered triples containing every ordered pair (a, b), $a \neq b$, in exactly one triple. Then the underlying sets of $\vec{T}$ satisfy* (M1) *and* (M2) *if and only if for every v the map $\sigma_v \colon b \mapsto c$, where $(v, b, c) \in \vec{T}$, is an $(n-1)$-cycle on $\{1, \ldots, n\} \setminus \{v\}$. The orientable neighbourly maps on n faces are exactly the sets of underlying sets of such $\vec{T}$.*

Proof. In both directions, σ_v is a permutation of the $n-1$ elements other than v, because each ordered pair (v, b) lies in exactly one member of $\vec{T}$.

Suppose every σ_v is an $(n-1)$-cycle. Then no σ_v has a 2-cycle, so no triple occurs in $\vec{T}$ with both of its cyclic orders: from (a, b, c) and (a, c, b) we would get $\sigma_a(b) = c$ and $\sigma_a(c) = b$. Hence the $n(n-1)/3$ underlying sets are distinct, and because the two ordered versions of an unordered pair lie in two different members of $\vec{T}$, every unordered pair lies in exactly two of those sets: that is (M1). The cycles of σ_v are the connected components of $\mathrm{lk}(v)$, so (M2) holds as well.

Conversely, suppose the underlying sets satisfy (M1) and (M2). Counting pair–triple incidences, (M1) forces exactly $n(n-1)/3$ underlying sets, so no triple occurs in $\vec{T}$ twice and no σ_v has a 2-cycle; the cycles of σ_v are then the components of $\mathrm{lk}(v)$, and (M2) says precisely that σ_v is an $(n-1)$-cycle.

Finally, a coherent orientation of the triangles of a triangulated closed surface induces on each edge the two opposite directions from its two incident triangles, which is exactly the condition that every ordered pair occurs once; conversely such a set of oriented triples orients the dual surface. So the orientable neighbourly maps are exactly those obtained this way. □

Lemma 3.6 (Framing count)**.** *Let $\mathcal{N}_n$ be the set of* orientable *neighbourly maps on the label set $\{0, 1, \ldots, n-1\}$ whose link at 0 is exactly the cycle $(1, 2, \ldots, n-1)$. Then*

$$|\mathcal{N}_n| \;=\; \sum_{C} \frac{2n(n-1)}{|\operatorname{Aut}(C)|},$$

the sum being over isomorphism classes C of orientable neighbourly maps on n faces. Moreover any isomorphism between two members of $\mathcal{N}_n$ is determined by the image of the label 0 together with the rotation or reflection it induces on the link of 0, so at most $2n(n-1)$ relabellings have to be tried to decide isomorphism.

Proof. Orientability is invariant under relabelling, so it is enough to argue inside one class. Fix a class C with representative M_0 and $a = |\operatorname{Aut}(C)|$. The labelled maps in C are the M_0^σ, $\sigma \in S_n$, and $M_0^\sigma = M_0^\tau$ exactly when $\sigma\tau^{-1} \in \mathrm{Aut}$. Because the map is neighbourly,

the link of a vertex v of M_0 consists of all other $n-1$ vertices, so a relabelling σ with $M_0^\sigma \in \mathcal{N}_n$ is determined by the vertex $v = \sigma^{-1}(0)$ — n choices — together with a bijection from $\mathrm{lk}(v)$ to $\{1, \dots, n-1\}$ carrying the link cycle of v to $(1, 2, \dots, n-1)$, of which there are exactly $2(n-1)$, one for each starting point and direction. Hence $2n(n-1)$ relabellings, and the members of $\mathcal{N}_n$ they produce are counted a times each. The last claim is the same computation read backwards. □

Remark 3.7. Non-orientable neighbourly maps on twelve faces also exist: [8, Theorem 4] enumerates triangular embeddings of K_{12} on the non-orientable surface of the same Euler characteristic. This is why orientability is part of the definition of $\mathcal{N}_n$. They are irrelevant here, because the surface of a polyhedron is orientable (Proposition 2.5).

Theorem 3.8 (Case list)**.** *Every neighbourly polyhedron with twelve faces satisfying* (H1)–(H3) *induces one of exactly* 59 *pairwise non-isomorphic orientable neighbourly maps on twelve faces.*

Proof. By Lemma 3.2 the polyhedron induces a neighbourly map T. Its dual is a triangulation with twelve vertices and complete edge graph K_{12}. It is orientable of genus 6 by Proposition 2.5, since it carries the dual cell structure of the same surface. The classification of [1], also reproduced in [8, Theorem 4], gives exactly 59 isomorphism classes of these orientable triangulations. Taking their combinatorial duals gives the claim. □

Recomputing the representatives. The implementation enumerates these maps without reading an external list. Lemma 3.5 reduces the search to cyclically oriented triples covering each ordered pair once. Fix the oriented link of one label to $(1, 2, \dots, 11)$, as allowed by Lemma 3.6; this fixes eleven of the forty-four triples. At each step choose an uncovered ordered pair (a, b) and try every third label c for which (b, c) and (c, a) are also uncovered. Reject an addition if it closes any partial link permutation into a cycle shorter than 11.

These rejections lose no completion: an ordered pair cannot be covered twice, and a closed proper link cycle cannot later be extended to a single 11-cycle. Conversely, every completed branch covers every ordered pair and has a single cycle at every label, so Lemma 3.5 makes it an orientable map. Every target map has a labelling with the fixed link and a unique coherent orientation extending that link; at each chosen pair its next triple is among the candidates. Induction therefore shows that the search visits every such map. At most forty-four triples are placed, so the search is finite. Canonicalization tries the 264 relabellings of Lemma 3.6. The implemented search returns 12 144 framed maps in 59 classes and checks that their canonical forms are exactly those of the shipped matrices. Appendix A gives the automorphism counts as reproducibility checks.

Identification with the published list. The formulas use the 59 matrices shipped with this project, in the project's own numbering. Their duals are also exactly the 59 triangulations listed by Lutz [14]. For each entry of that list, the accompanying data give a bijection of the twelve labels taking one shipped map to that entry; every map is used once. Applying these bijections reproduces the published facet lists, whose normalized SHA-256 is recorded with the comparison. This check uses the explicit bijections, not the enumeration's canonicalization algorithm. The Manifold Page cites [20, 8] for its twelve-vertex enumerations. No external list is needed to generate the formulas or to rerun the census.

Thus the classification supplies the completeness theorem, the search supplies a reproducible recomputation, and the explicit correspondence identifies the concrete input data with a published list. Agreement of counts alone would not identify the maps. The orbit identity and the small-size controls test the implementation; they are not the completeness argument. The classification itself is an existing computational classification, not a new classification proved by the SAT computation proposed here.

4. General position

Everything that follows reads a realization through the signs of its brackets, which requires those signs to be non-zero.

Hypothesis 4.1 (General position)**.**

(G) $[S] \neq 0$ for every 4-subset $S \subseteq \{0, 1, \dots, n\}$.

Geometrically: no four of the $n+1$ covectors are linearly dependent. The brackets containing 0 say that any three of the planes $h_1, \dots, h_n$ meet in exactly one affine point, because $[0, i, j, k] = -\det(n_i, n_j, n_k)$ by Lemma 5.1; the brackets not containing 0 say that no four of the planes share a point of projective space — no four pass through a common affine point, and no four are parallel to a common direction. For $n = 12$ that is 220 conditions of the first kind and 495 of the second.

Theorem 4.2 (Reduction to general position)**.** *Every neighbourly map admitting a realization satisfying* (R1)–(R4) *also admits one satisfying* (G)*.*

The main point is to preserve the embedded surface when its supporting planes move. We use an auxiliary triangulation and the following elementary stability fact; compare [13, Section 2, Lemmas 2.1–2.2]. The triangulation is used only in this proof and does not change the faces of the map or the Boolean encoding.

Lemma 4.3 (Stability of a finite triangulation)**.** *Let K be a finite geometric simplicial complex in $\mathbb{R}^d$. Every sufficiently small displacement of its vertices extends, affinely on each simplex, to an embedding $F\colon |K| \to \mathbb{R}^d$.*

Proof. First, such a map is injective if the images of every two simplices with disjoint vertex sets are disjoint. To see this, suppose $F(x) = F(y)$ for distinct $x, y \in |K|$. Write their barycentric coordinates as (α_v) and (β_v) over the vertices of K, with zero coefficients outside their supporting simplices. The coefficients $c_v = \alpha_v - \beta_v$ are not all zero and satisfy

$$\sum_v c_v = 0, \qquad \sum_v c_v F(v) = 0.$$

The positive coefficients are supported on a face of the simplex supporting x, and the negative coefficients on a face of the simplex supporting y. These two faces have disjoint vertex sets. Dividing by $\sum_{c_v>0} c_v = -\sum_{c_v<0} c_v > 0$ gives a point in both images, a contradiction.

In the original geometric complex, any two simplices with disjoint vertex sets are disjoint compact sets, hence at positive distance. If there are such pairs, let $\delta > 0$ be the minimum of their finitely many distances. Move every vertex by less than $\delta/3$. A point of any simplex, expressed as a convex combination of its vertices, moves by less than $\delta/3$; therefore the images of each such pair remain disjoint. The criterion above proves injectivity. If there are no such pairs, the criterion applies without a distance bound. The affine pieces agree on common faces, so F is continuous; compactness of $|K|$ makes it an embedding. □

Proof of Theorem 4.2. Fix a realization $q^* = (q_1^*, \dots, q_n^*)$. Triangulate each of its simple polygonal faces by diagonals, using its existing vertices and nondegenerate triangles. Such a triangulation exists also for nonconvex simple polygons [15]. By (R3), the triangulations fit together along the original edges to form a finite geometric simplicial complex K: two faces $i \neq j$ meet only in the segment $E(i, j)$, an edge of both polygons that no triangulation subdivides, and a triangle of either face meets that segment in the empty set, in one endpoint, or in the whole segment as one of its edges; so a triangle of face i and a triangle of face j meet in a common face. Let K_f be the subcomplex triangulating face f.

For q close to q^*, move each original vertex $v = \{i, j, k\} \in T$ to

$$p_v(q) = P_{ijk}(q) = h_i(q) \cap h_j(q) \cap h_k(q).$$

These points exist and depend continuously on q: the incident normal determinants are nonzero at q^* by (R1), and Cramer's rule applies in a neighbourhood. There are finitely many vertices, so Lemma 4.3 shows that the resulting affine map F_q on K is an embedding for all sufficiently close q.

Every vertex of K_f lies in $h_f(q)$ after the displacement. Thus $F_q(|K_f|)$ is an embedded planar disk with the prescribed boundary polygon, and equals the closed region bounded by that polygon. Indeed, F_q maps the boundary circle of $|K_f|$ injectively onto that polygon, which is therefore simple, and by invariance of domain it maps the interior of $|K_f|$ onto an open subset of the plane $h_f(q)$; this set is connected, bounded and closed in the complement of the polygon, so it is the bounded component of that complement. In particular, (R2) holds. Injectivity gives, for $i \neq j$,

$$F_q(|K_i|) \cap F_q(|K_j|) = F_q\big(|K_i| \cap |K_j|\big) = E(i, j)(q),$$

which is (R3). The incident normal determinants remain nonzero, and distinct vertices remain distinct by injectivity, giving (R1). The normals remain nonzero and the supporting planes remain pairwise distinct by continuity, giving (R4). Hence a whole neighbourhood of q^* in $Q = (\mathbb{R}^4)^n$ consists of realizations of the same map.

For each 4-subset $S \subseteq \{0, 1, \ldots, n\}$, the bracket $[S]$ is a nonzero polynomial on Q, with $q_0 = (0, 0, 0, 1)$ fixed. Indeed, its value can be made nonzero by choosing the four rows to be a basis of $\mathbb{R}^4$; when $0 \in S$, choose the other rows to be e_1, e_2, e_3. The product of these finitely many polynomials is nonzero, so its zero set contains no open ball. The neighbourhood just obtained therefore contains a realization at which all brackets are nonzero, as required. □

Remark 4.4. The argument also shows that the new realization can be chosen arbitrarily close to the original one. It moves the supporting planes and derives the vertices from their intersections; arbitrary independent vertex displacements would not preserve the planarity of the original faces.

We record two local geometric facts used in the sign dictionary below; in them q^* is any realization of T.

Lemma 4.5 (Nondegeneracy is open)**.** *At q^* the normals $n(q_i), n(q_j), n(q_k)$ are independent for every $\{i, j, k\} \in T$ and $n(q_i), n(q_j)$ are independent for every $i \neq j$. Consequently, on a neighbourhood of q^* the vertex maps $q \mapsto P_{ijk}(q)$ and the line maps $q \mapsto L_{ij}(q)$ are real-analytic, and all vertices stay inside one fixed ball.*

Proof. Independence at a vertex is (R1). For a pair, if $n(q_i)$ and $n(q_j)$ were parallel the planes would be equal, excluded by (R4), or disjoint, and then $E(i, j) \subseteq h_i \cap h_j$ could not be a segment. Both conditions are the non-vanishing of a determinant, hence open, and on the set where they hold Cramer's rule expresses P_{ijk} and a direction vector and base point of L_{ij} as rational functions of q with non-vanishing denominators. □

Lemma 4.6 (No straight angles)**.** *At q^* no three consecutive vertices of a face polygon are collinear. Hence on a neighbourhood the turn at every corner is non-zero and of constant sign.*

Proof. Three consecutive vertices of the polygon of f are $P_{f,a,x}$, $P_{f,a,b}$, $P_{f,b,y}$ with a, b consecutive in $\mathrm{lk}(f)$. The first two lie on L_{fa} and are distinct by (R1); the last two lie on L_{fb} and are distinct. If the three were collinear, the line through them would equal both L_{fa} and L_{fb}, so h_f, h_a, h_b would share a line and their normals would be dependent, contradicting Lemma 4.5. The turn at a corner is a continuous function of q which is non-zero at q^*, hence of constant sign nearby. □

Convention 4.7. From now on all realizations are assumed to satisfy (G). By Theorem 4.2 this loses no map: if some map is realizable at all, it is realizable in general position.

5. The sign dictionary

Throughout this section q is a realization of a neighbourly map T on n faces satisfying (G). For a covector q_u write $h_u(x) = q_u \cdot (x, 1)$ for the affine form whose zero set is the plane, and $n_u = n(q_u)$ for its normal. For S a 4-subset of $\{0, 1, \dots, n\}$ put

$$X_S = \begin{cases} 1, & [S] < 0, \\ 0, & [S] > 0, \end{cases}$$

which is defined by (G); for an ordered tuple, X changes by the parity of the permutation, so $X_{\sigma(S)} = X_S \oplus \mathrm{par}(\sigma)$. All the identities below are exact, and all of them express a geometric predicate as an $\mathbb{F}_2$-linear or small Boolean function of the $\binom{n+1}{4}$ bits X_S.

5.1. Two cofactor identities.

Lemma 5.1. *For distinct $i, j, k, l \in \{1, \dots, n\}$ with n_i, n_j, n_k independent,*

$$[0, i, j, k] = -\det(n_i, n_j, n_k), \qquad [i, j, k, l] = \det(n_i, n_j, n_k)\, h_l(P_{ijk}).$$

Proof. The first is the expansion of the determinant along the row $q_0 = (0, 0, 0, 1)$, whose only non-zero entry sits in the last column with sign $(-1)^{1+4}$. For the second, change affine coordinates by $y = x - P_{ijk}$. A covector $q_u = (n_u, d_u)$ becomes $(n_u, h_u(P_{ijk}))$, because $q_u \cdot (x, 1) = n_u \cdot y + h_u(P_{ijk})$; this is right multiplication of the row by a unipotent matrix, so the 4×4 determinant is unchanged. In the new coordinates the first three rows have last entry 0, and expanding along the last column leaves $h_l(P_{ijk}) \cdot \det(n_i, n_j, n_k)$. □

5.2. Order along a line.

Fix $i \neq j$ and orient the line L_{ij} by the direction $d_{ij} = n_i \times n_j$, which is non-zero by Lemma 4.5. The other $n - 2$ planes mark $n - 2$ points P_{ijk} on it, pairwise distinct by (G). Let $\mathrm{Ord}(i, j; k, l) = 1$ when P_{ijk} precedes P_{ijl} in that direction, and 0 otherwise.

Theorem 5.2 (Order bits)**.** *For all distinct $i, j, k, l \in \{1, \dots, n\}$,*

$$\mathrm{Ord}(i, j; k, l) \;=\; X_{0ijk} \oplus X_{0ijl} \oplus X_{ijkl}.$$

Proof. Write $D_{ijk} = \det(n_i, n_j, n_k)$ and $P_{ijl} - P_{ijk} = t\, d_{ij}$; then $\mathrm{Ord}(i, j; k, l) = [t > 0]$. Since $h_l(P_{ijl}) = 0$ and $n_l \cdot d_{ij} = \det(n_l, n_i, n_j) = D_{ijl}$,

$$h_l(P_{ijk}) = h_l\big(P_{ijl} - t\, d_{ij}\big) = -t\, D_{ijl},$$

so Lemma 5.1 gives $[i, j, k, l] = -t\, D_{ijk} D_{ijl}$ and $[0, i, j, k] = -D_{ijk}$, $[0, i, j, l] = -D_{ijl}$. All factors are non-zero by (G). Writing $\alpha = [D_{ijk} < 0]$, $\beta = [D_{ijl} < 0]$ and $\gamma = [t < 0]$ we get $X_{0ijk} = 1 \oplus \alpha$, $X_{0ijl} = 1 \oplus \beta$ and $X_{ijkl} = (1 \oplus \gamma) \oplus \alpha \oplus \beta$, because the sign bit of a product is the sum of the sign bits. Adding the three gives $1 \oplus \gamma = [t > 0]$. □

Remark 5.3 (Why a 0 appears)**.** A projective line carries no linear order; what a rank-4 chirotope gives on L_{ij} is a *circular* order of the marked points. Removing one point turns it into a linear order, and the point removed in Theorem 5.2 is the line's own point at infinity, P_{0ij}. This is why the element 0 occurs in the formula, and recognizing it is what makes Layer A of Section 6 possible.

5.3. **Betweenness inside an edge.**

Definition 5.4. For $i \neq j$ with $\mathrm{Ends}(i,j) = \{z_0, z_1\}$ and $u \notin \{i, j, z_0, z_1\}$ put

$$\mathrm{OnB}(i,j;u) \;=\; \mathrm{Ord}(i,j;z_0,u) \oplus \mathrm{Ord}(i,j;z_1,u).$$

Lemma 5.5. $\mathrm{OnB}(i,j;u) = 1$ *if and only if* P_{iju} *lies in the relative interior of the edge* $E(i,j)$.

Proof. $E(i,j)$ is the segment of L_{ij} between P_{ijz_0} and P_{ijz_1}. In a strict total order, exactly one of the two endpoints precedes a given point precisely when that point lies strictly between them; by (G) the point P_{iju} coincides with neither endpoint, since that would make four planes concurrent. □

Lemma 5.6 (Cyclic adjacency)**.** *Two edges* $E(f,a)$ *and* $E(f,b)$ *of the face* f *are consecutive in the boundary polygon of* F^f *if and only if* $\{f,a,b\} \in T$*, and then they meet exactly at the vertex* P_{fab}.

Proof. By construction the boundary polygon of F^f visits the faces of $\mathrm{lk}(f)$ in the cyclic order of that link, and a, b are adjacent in $\mathrm{lk}(f)$ exactly when $\{f,a,b\} \in T$ (Definition 3.1). Consecutive edges meet at the corresponding vertex, and only there by simplicity of the polygon. □

5.4. **Turns, orientation and reflex corners.** Let a, b be consecutive in $\mathrm{lk}(f)$, in the direction of a fixed traversal of the link cycle, and let $v = P_{fab}$ be the corner between the edges $E(f,a)$ and $E(f,b)$. Let x be the third face at the other endpoint of $E(f,a)$ and y the third face at the other endpoint of $E(f,b)$, that is $\mathrm{Ends}(f,a) = \{b,x\}$ and $\mathrm{Ends}(f,b) = \{a,y\}$. Put

$$u_{\mathrm{in}} = v - P_{fax}, \qquad u_{\mathrm{out}} = P_{fby} - v, \qquad w(f;a,b) = \big(u_{\mathrm{in}} \times u_{\mathrm{out}}\big) \cdot n_f.$$

By Lemma 4.6, $w(f;a,b) \neq 0$.

Lemma 5.7 (Turn bits)**.** $\big[w(f;a,b) < 0\big] \;=\; 1 \oplus \mathrm{Ord}(f,a;x,b) \oplus \mathrm{Ord}(f,b;a,y) \oplus X_{0fab}$.

Proof. Orient L_{fa} by $d_{fa} = n_f \times n_a$, the direction used in Theorem 5.2. Then $u_{\mathrm{in}} = s\, d_{fa}$ with $[s < 0] = 1 \oplus \mathrm{Ord}(f,a;x,b)$, because $\mathrm{Ord}(f,a;x,b) = [P_{fax} < P_{fab}]$ and $u_{\mathrm{in}} = P_{fab} - P_{fax}$. Likewise $u_{\mathrm{out}} = t\, d_{fb}$ with $[t < 0] = 1 \oplus \mathrm{Ord}(f,b;a,y)$. From the identity $(A \times B) \times (A \times C) = \det(A,B,C)\, A$,

$$\big(d_{fa} \times d_{fb}\big) \cdot n_f = \big((n_f \times n_a) \times (n_f \times n_b)\big) \cdot n_f = \det(n_f, n_a, n_b)\, \|n_f\|^2,$$

and $\det(n_f, n_a, n_b) = -[0,f,a,b]$ by Lemma 5.1, so $[\det(n_f,n_a,n_b) < 0] = 1 \oplus X_{0fab}$. Since $w = s\,t\,(d_{fa} \times d_{fb}) \cdot n_f$ and the sign bit of a product is the sum of the sign bits,

$$[w < 0] = (1 \oplus \mathrm{Ord}(f,a;x,b)) \oplus (1 \oplus \mathrm{Ord}(f,b;a,y)) \oplus (1 \oplus X_{0fab}),$$

which is the claim. □

Lemma 5.8 (One orientation bit per face)**.** *For each face* f *there is* $s_f \in \mathbb{F}_2$*, depending on* f *and the chosen traversal only, such that for every corner* (a,b) *of* f

$$\mathrm{refl}(f;a,b) \;:=\; \big[w(f;a,b) < 0\big] \oplus s_f$$

equals 1 *exactly when the interior angle of* F^f *at* P_{fab} *exceeds* π*. We write* $\mathrm{refl}(f;a,b)$ *also when the pair is given in the opposite order, meaning the same bit: the pair is normalized to link-cycle order before the formula of Lemma 5.7 is applied.*

Proof. Orient the plane h_f by n_f. A simple closed polygon in an oriented plane is traversed either positively or negatively; let $s_f = 0$ in the first case and $s_f = 1$ in the second. For a positively traversed simple polygon the interior angle at a corner exceeds π exactly when the turn from the incoming to the outgoing direction is negative, that is when $w < 0$;

for a negatively traversed one the condition is $w > 0$. Both cases are summarized by $[w < 0] \oplus s_f$. The bit s_f does not depend on the corner, because the traversal direction is a property of the polygon. □

Remark 5.9 (An easy way to get this wrong)**.** Both sides of Lemma 5.7 change under exchanging a and b: the corner is the same, but the formula computes the complement. The predicate "this corner is reflex" is symmetric while its expression is antisymmetric, and the two are reconciled only by always writing the pair in the direction of the link traversal. An implementation that reaches a corner in the other order produces a well-formed formula that encodes something else.

Lemma 5.10 (Crossing parity fixes the orientation)**.** *Let f be a face, $g \in \mathrm{lk}(f)$, $\mathrm{Ends}(f, g) = \{z_0, z_1\}$ and $V = P_{f,g,z_1}$. For $v \notin \{f, g, z_0, z_1\}$ put*

$$C_v = \mathrm{OnB}(f, v; g), \qquad B_v = \mathrm{Ord}(f, g; z_1, z_0) \oplus \mathrm{Ord}(f, g; z_1, v).$$

Then the corner of F^f at V is reflex if and only if $\bigoplus_v C_v \wedge B_v = 1$.

Proof. Let R be the open ray of L_{fg} issuing from V in the direction opposite to the edge $E(f, g)$; thus $B_v = 1$ says exactly that the point P_{fgv}, which is where the line L_{fv} meets L_{fg}, lies on R: indeed B_v is the exclusive or of $[V < P_{fgz_0}]$ and $[V < P_{fgv}]$, which says that P_{fgv} and the other endpoint P_{fgz_0} of $E(f, g)$ lie on opposite sides of V.

A plane wedge contains the ray opposite to each of its bounding rays if and only if its angle exceeds π. Near V the region F^f is the wedge bounded by the germs of $E(f, g)$ and $E(f, z_1)$, and R is opposite to the first of those germs, so a point $V + \varepsilon\, d \in R$ with $\varepsilon > 0$ small lies in F^f exactly when the corner at V is reflex.

Now follow R from V to infinity. Far enough out it leaves the bounded region F^f; membership in F^f changes exactly at transverse crossings of ∂F^f, and by (G) the ray meets no vertex of the polygon — a vertex of ∂F^f on L_{fg} would make four planes concurrent — and meets no edge non-transversally except the edge $E(f, g)$ itself, which lies on L_{fg} and on the far side of V. Hence the membership bit at $V + \varepsilon d$ is the parity of the number of edges of ∂F^f that R crosses.

Which edges are those? Edge $E(f, g)$ is behind V. Edge $E(f, z_1)$ has V as an endpoint and its supporting line meets L_{fg} only at V, which is not on the open ray. Edge $E(f, z_0)$ has the other endpoint P_{fgz_0} of $E(f, g)$ as an endpoint, and its supporting line meets L_{fg} exactly there, again not on R. For every other v, the edge $E(f, v)$ lies on L_{fv}, which is not parallel to L_{fg} — that would make h_f, h_g, h_v share a direction and $[0, f, g, v]$ vanish, against (G) — and therefore meets L_{fg} in exactly the point P_{fgv}, and that point belongs to R exactly when $B_v = 1$ and to the relative interior of $E(f, v)$ exactly when $C_v = 1$ (Lemma 5.5). At such a point the two supporting lines are distinct, so the intersection is transverse; by (G) it lies in the interior of a single boundary edge, not at a polygon vertex. Thus R crosses $E(f, v)$ if and only if $C_v \wedge B_v = 1$, and the parity of the crossings is $\bigoplus_v C_v \wedge B_v$. □

Remark 5.11. Lemma 5.10 is what removes the last free bit from the encoding. By Lemma 5.8 the reflexness of a corner is a bracket expression *up to* the unknown s_f; Lemma 5.10 computes one particular corner's reflexness from brackets alone, which determines s_f and hence all eleven corners of that face. Without it, a formula could satisfy every other layer by flipping a face's orientation bit, and the encoded condition would be strictly weaker.

6. The chirotope layers S, H and A

6.1. **Contractions and the cut order.** For a pair $\{i, j\} \subseteq \{0, 1, \ldots, n\}$ let $\chi_{ij}(x, y) = X_{ijxy}$ on the remaining $n-1$ elements. Each χ_{ij} is a *tournament*: $\chi_{ij}(y, x) = 1 \oplus \chi_{ij}(x, y)$.

There are $\binom{n+1}{2} = 78$ such contractions for $n = 12$. The three-term Grassmann–Plücker sign condition for the rank-4 bracket function on a 4-subset $\{a, b, c, d\}$ of the complement of $\{i, j\}$ is exactly the rank-2 condition for χ_{ij} on that subset:

> **(GP)** the three products $[ijab][ijcd]$, $-[ijac][ijbd]$, $[ijad][ijbc]$ do not all have the same sign.

This is the content of the Plücker relation $[ijab][ijcd] - [ijac][ijbd] + [ijad][ijbc] = 0$: three non-zero reals summing to zero cannot share a sign.

Lemma 6.1 (Cut order)**.** *Let χ be a tournament on a finite set X with $|X| \geq 4$, let $r \in X$, and define for $x, y \in X \setminus \{r\}$*

$$d(x, y) \;=\; \chi(x, y) \oplus \chi(r, x) \oplus \chi(r, y).$$

Then χ satisfies (GP) *on every 4-subset of X if and only if d is a strict total order on $X \setminus \{r\}$.*

Proof. Reorient every $x \neq r$ by $t_x = 1 \oplus \chi(r, x)$, leaving r unchanged; that is, replace χ by $\chi'(u, v) = \chi(u, v) \oplus t_u \oplus t_v$ with $t_r = 0$. Then $\chi'(r, x) = 1$ for all $x \neq r$ and $\chi'(x, y) = d(x, y)$. Reorientation multiplies each of the three products in (GP) by the same sign, so it preserves (GP).

Suppose (GP) holds. Apply it to a 4-subset $\{r, x, y, z\}$: the sign bits of the three products are $1 \oplus d(y, z)$, $d(x, z)$ and $1 \oplus d(x, y)$, and they are all equal exactly when $d(x, y) = d(y, z) = 1 \oplus d(x, z)$, which is precisely one of the two directed 3-cycles on $\{x, y, z\}$. Hence d has no directed 3-cycle. A tournament without directed 3-cycles is transitive — if $x < y$ and $y < z$ but not $x < z$, then x, y, z form one — so d is a strict total order.

Conversely, suppose d is a strict total order. Choose reals t_x strictly decreasing along it and put $v_r = (0, 1)$ and $v_x = (1, t_x)$ in $\mathbb{R}^2$. Then $\det(v_r, v_x) = -1 < 0$ and $\det(v_x, v_y) = t_y - t_x$, so the negativity bits of these 2×2 determinants are exactly χ'. Two-by-two determinants of vectors in the plane satisfy the Plücker identity, so (GP) holds for χ' on every 4-subset, hence for χ. □

Remark 6.2 (Why transitivity of the raw tournament would be wrong)**.** It is tempting to impose transitivity on χ_{ij} itself. That is false: eleven vectors in a plane have a *circular* order, and three at 120° form a directed 3-cycle. Among the $2^{\binom{m}{2}}$ tournaments on m elements, exactly $(m-1)!\,2^{m-1}$ satisfy (GP) on all 4-subsets while only $m!$ are transitive; the ratio is $2^{m-1}/m$, which at $m = 11$ is $1024/11 \approx 93$. Imposing transitivity would delete realizations. Cutting the circle at one element — passing from χ to d — is what turns the circular order into a linear one, and Lemma 6.1 says nothing is lost or gained by doing so.

6.2. **The three layers.** Fix a neighbourly map T on $n = 12$ faces. The formula has one Boolean variable X_S for each of the $\binom{13}{4} = 715$ brackets, together with the auxiliary variables introduced below.

Layer S (definitions):

> For each of the 66 pairs $\{i, j\} \subseteq \{1, \ldots, 12\}$ and each pair $\{k, l\}$ of the ten remaining faces, a variable o^{ij}_{kl} with the defining clauses of
>
> $$o^{ij}_{kl} \;=\; X_{0ijk} \oplus X_{0ijl} \oplus X_{ijkl}.$$
>
> By Theorem 5.2 this is the order of P_{ijk} and P_{ijl} along L_{ij}; by Remark 5.3 it is the cut order of the contraction χ_{ij} with cut point 0. Being a definition, this layer excludes no assignment of the bracket bits.

Layer H (transitivity on the 66 affine lines):

> For each such pair and each triple k, l, m of the remaining faces, the two clauses expressing $o^{ij}_{kl} \wedge o^{ij}_{lm} \to o^{ij}_{km}$ and its converse orientation.

Layer A (transitivity on the 12 lines at infinity):

For each face b let r_b be the smallest face other than b. For each pair $\{x, y\}$ of the ten faces other than b, r_b a variable

$$d^b_{xy} \;=\; X_{0bxy} \oplus X_{0br_bx} \oplus X_{0br_by},$$

and for each triple the two transitivity clauses.

Layer A needs the separate treatment because for a pair $\{0, b\}$ the element 0 is already one of the two defining labels and cannot serve as the cut point; a different cut point r_b is chosen. Geometrically, $[0, b, x, y] = -\det(n_b, n_x, n_y)$ by Lemma 5.1, so the contraction at $\{0, b\}$ is the chirotope of the eleven other normals modulo n_b: eleven vectors in a plane, and d^b is their cut order.

Theorem 6.3 (What S, H and A say)**.** *An assignment of the* 715 *bits X_S extends to a satisfying assignment of $S \wedge H \wedge A$ if and only if the map $S \mapsto (-1)^{X_S}$, extended by alternation and set to zero on tuples with repeated labels, is a uniform chirotope of rank* 4 *on the* 13 *elements $\{0, 1, \ldots, 12\}$. Moreover every realization in general position gives such an assignment.*

Proof. The auxiliary variables of S and A are defined by their layers, so an assignment of the X_S extends in exactly one way; what H and A then require is that, for each of the 78 contractions, the cut order derived from χ_{ij} — with cut point 0 for the 66 pairs of finite labels and cut point r_b for the 12 pairs containing 0 — is a strict total order. By Lemma 6.1 this holds for a given contraction if and only if that contraction satisfies (GP) on all its 4-subsets. Ranging over the contractions, $S \wedge H \wedge A$ is satisfiable over a given bracket assignment if and only if all three-term Grassmann–Plücker sign conditions hold. The alternating extension has every 4-subset as a basis, so its support is the uniform matroid $U_{4,13}$. This supplies the matroid-support hypothesis (B1′) of [5, Theorem 3.6.2, p. 138]; it must not be omitted for sign maps with zeros. With $(x_1, x_2, x_3, x_4, y_1, y_2) = (a, b, i, j, c, d)$, condition (B2″) of that theorem excludes the all-negative triple of products in (GP), and interchanging a, b excludes the all-positive triple. Conversely, (GP) implies this condition; repeated-label instances follow from alternation. Thus the theorem identifies precisely these conditions with the rank-4 chirotope axioms, proving the first claim.

For the second, let q be a realization in general position. Its brackets are the 4×4 minors of the 13×4 matrix of covectors, all non-zero by (G), so they form a uniform rank-4 chirotope and satisfy every Plücker relation. Hence each contraction satisfies (GP), and Lemma 6.1 gives a strict total order for every chosen cut point. For the finite pairs this is the order of Theorem 5.2, satisfying S and H; for the pairs $\{0, b\}$ it is the cut order d^b, satisfying A. This necessary direction uses only determinant identities and Lemma 6.1, not the abstract chirotope characterization of [5]. □

Remark 6.4. The relational form of the same condition — one clause of width six per Plücker relation and per sign pattern — would need 411 840 clauses for $n = 12$. The formulation through 78 cut orders uses 46 800 clauses of width three and four (Appendix A). This reduces the clause count; its effect on solving time requires measurement. The broader use of SAT to study oriented-matroid realizability has precedents in [19, 18]. Those references motivate the SAT approach; the cut-order encoding used here is established by the preceding lemmas.

7. The geometric layers B and I

7.1. **Layer B: face boundaries are simple.** For a face f and two faces a, b that are *not* adjacent in $\mathrm{lk}(f)$ — equivalently, by Lemma 5.6, with $\{f, a, b\} \notin T$ — the edges

$E(f,a)$ and $E(f,b)$ are non-consecutive edges of the boundary polygon of F^f, and their supporting lines meet at the single point P_{fab}. Layer B consists of the clauses

$$\neg\,\mathrm{OnB}(f,a;b)\ \vee\ \neg\,\mathrm{OnB}(f,b;a)$$

for all such triples, where OnB is as in Definition 5.4, together with the defining clauses of the OnB variables.

Lemma 7.1. *Every realization in general position satisfies Layer B.*

Proof. If both bits were 1 then, by Lemma 5.5, P_{fab} would be interior to both $E(f,a)$ and $E(f,b)$, so two non-consecutive closed edges of the boundary polygon of F^f would meet, and the polygon would not be simple, contradicting (R2). □

7.2. **Layer I: orientation and reflex corners.** Layer I has two parts. For each face f it contains the defining clauses of the corner bits $\mathrm{refl}(f;a,b)$ of Lemma 5.8 — one auxiliary variable s_f per face and, per corner, one turn variable and the one helper variable that splitting its four-term exclusive-or introduces (Appendix A) — together with the *orientation equation* of Lemma 5.10,

$$\mathrm{refl}(f;g,z_1)\ =\ \bigoplus_{v\notin\{f,g,z_0,z_1\}}\big(C_v\wedge B_v\big),$$

where g is the first face of $\mathrm{lk}(f)$ and $\{z_0,z_1\}=\mathrm{Ends}(f,g)$. Either element of $\mathrm{Ends}(f,g)$ may play the role of z_1 — Lemma 5.10 holds for both, with B_v read relative to the chosen one — while the corner bit on the left, being attached to the corner at P_{f,g,z_1}, is written with its pair normalized to link-cycle order as in Lemma 5.8 and Remark 5.9. The two choices give different clause sets, so one is fixed once and for all, like the emission order, and belongs to the formula's identity; the implementation takes z_1 to be the larger of the two labels. Layer I contains further, for each of the 44 vertices $\{f,g,h\}\in T$, the three clauses

$$\neg\mathrm{refl}(f;g,h)\vee\neg\mathrm{refl}(g;f,h),\quad \neg\mathrm{refl}(f;g,h)\vee\neg\mathrm{refl}(h;f,g),\quad \neg\mathrm{refl}(g;f,h)\vee\neg\mathrm{refl}(h;f,g),$$

each corner bit being written with its pair in link-cycle order as in Remark 5.9.

Lemma 7.2. *Every realization in general position satisfies Layer I, with s_f given by Lemma 5.8.*

Proof. With s_f the traversal bit of Lemma 5.8, the variable $\mathrm{refl}(f;a,b)$ takes the value "the corner of F^f at P_{fab} is reflex". The orientation equation then holds by Lemma 5.10, and the three clauses at each vertex hold by Lemma 2.6. □

Remark 7.3. The vertex clauses couple the corners of three different faces. They encode the local intersection restriction of Lemma 2.6; the other geometric clauses concern one face polygon at a time.

8. The gauge

The bracket bits of a realization are not unique: two operations change them without changing the polyhedron.

Lemma 8.1 (The reorientation group)**.** *Let $G=\mathbb{F}_2^{n+1}$ with coordinates $(\gamma,\epsilon_1,\dots,\epsilon_n)$ act on sign vectors by*

$$X_S\ \longmapsto\ X_S\ \oplus\ \Big(\gamma\ \oplus\ \bigoplus_{i\in S,\ i\geq 1}\epsilon_i\Big).$$

Then G maps the bracket vector of a realization to the bracket vector of a realization of the same map, and the $\binom{n+1}{4}$ functionals $\varphi_S(\gamma,\epsilon)=\gamma\oplus\bigoplus_{i\in S,i\geq 1}\epsilon_i$ span the dual of G.

Proof. Replacing q_i by $-q_i$ changes no plane, hence no realization, and flips exactly the brackets containing i: this is ϵ_i. For γ, let $\tau(x) = Ax + b$ be an orientation-reversing affine map of $\mathbb{R}^3$ and $M = \left(\begin{smallmatrix} A & b \\ 0 & 1 \end{smallmatrix}\right)$, so that a covector q is carried to qM^{-1}. The image of a realization is a realization of the same map, since τ is a homeomorphism of $\mathbb{R}^3$ carrying planes to planes. All four rows transform by the same matrix, so every bracket is multiplied by $\det(M^{-1}) = 1/\det A < 0$ and every bit flips. Moreover the last row of M^{-1} is $(0, 0, 0, 1)$, so $q_0 M^{-1} = q_0$: the plane at infinity is fixed, as the action above assumes.

For the span, suppose a vector (γ, ϵ) is annihilated by every φ_S, i.e. $\varphi_S(\gamma, \epsilon) = 0$ for all S. Comparing $S = \{i, j, k, l\}$ with $i, j, k, l \geq 1$ and $S' = \{0, j, k, l\}$, which exists because $n \geq 4$, gives $\epsilon_i = 0$; as i was arbitrary all ϵ_i vanish, and then $\gamma = \varphi_S(\gamma, 0) = 0$. Hence the functionals span the dual. □

Corollary 8.2 (Gauge)**.** *There are brackets S_k, for $1 \leq k \leq n + 1$, whose functionals φ_{S_k} form a basis of the dual of G. The action of G on their sign patterns is simply transitive. Thus every realization can be moved by a unique element of G to one in which all these brackets are positive. Adding a unit clause $\neg X_{S_k}$ for each $k = 1, \ldots, n + 1$ therefore preserves the property that a realization yields a satisfying assignment.*

Proof. A spanning set contains a basis, which gives the S_k; a basis of the dual of a finite-dimensional $\mathbb{F}_2$-vector space makes the action of G on the vector of values simply transitive. Given a realization, apply the unique group element taking its pattern on $S_1, \ldots, S_{n+1}$ to 0; by Lemma 8.1 the result is again a realization of the same map, now satisfying the unit clauses. □

Remark 8.3. The gauge is a symmetry quotient, not a geometric restriction: it removes 2^{n+1}-fold redundancy from the search space. In the implementation the basis is chosen greedily among the brackets that contain a designated *gauge face* φ, completed by one bracket that does not contain it. The completion is necessary, not cosmetic: the functionals of the $\binom{n}{3}$ brackets through a fixed element φ all annihilate $(\gamma, \epsilon) = (1, \epsilon_\varphi{=}1, 0, \ldots, 0)$, so they span only an n-dimensional space and can never form a basis. Twelve of the thirteen pinned brackets may contain φ; the thirteenth may not. Which basis is chosen affects the shape of the search but not the theorem. The gauge is recorded as part of the formula's identity, because a formula with a different gauge is a different formula.

9. The formula and the reduction

Definition 9.1 (The formula)**.** For a neighbourly map T on $n = 12$ faces and a gauge face φ, let Φ_T — the gauge face being fixed once and for all and suppressed from the notation — be the CNF formula in the variables

$$\underbrace{X_S}_{715}, \quad \underbrace{o^{ij}_{kl}}_{2\,970}, \quad \underbrace{d^b_{xy}}_{540}, \quad \underbrace{\mathrm{OnB}(i, j; u)}_{528}, \quad \underbrace{s_f,\ \text{turn and parity auxiliaries}}_{504}$$

consisting of the layers S, H, A (Section 6), B and I (Section 7) emitted in that order, followed by the 13 gauge clauses of Corollary 8.2. Its size is

layer	clauses	variables
brackets	0	715
S	23 760	2 970
H	15 840	0
A	7 200	540
B	2 640	528
I	2 724	504
gauge	13	0
Φ_T	52 177	5 257

The emission order is part of the formula's identity: it fixes the numbering of the variables, and a case or a cube in Section 9.1 names a region of the search space only relative to a fixed numbering.

Theorem 9.2 (Necessity)**.** *Let T be a neighbourly map on twelve faces and let q be a realization of T. Then Φ_T is satisfiable.*

Proof. By Theorem 4.2 we may assume q satisfies (G), and by Corollary 8.2 we may assume in addition that the 13 gauge brackets are positive, both replacements producing another realization of the same T. Assign to X_S the negativity bit of $[S]$, to o^{ij}_{kl} and d^{b}_{xy} the values forced by their defining clauses, to $\mathrm{OnB}(i, j; u)$ the value forced by Definition 5.4, to s_f the traversal bit of Lemma 5.8 and to each turn and parity auxiliary the value forced by its definition. Layer S and the definitional clauses of A, B and I hold by construction; layers H and A hold by Theorem 6.3; layer B by Lemma 7.1; layer I by Lemma 7.2; the gauge clauses by the choice of representative. □

Corollary 9.3 (Conditional non-existence)**.** *Suppose that for each of the* 59 *neighbourly maps T of Theorem 3.8 the formula Φ_T is unsatisfiable. Then there is no neighbourly polyhedron with twelve faces satisfying* (H1)–(H3)*; that is, the Question of Section 1 has a negative answer within these conventions.*

Proof. Such a polyhedron has trivalent vertices (Lemma 2.4) and its induced map is isomorphic to one of the 59 representatives T (Theorem 3.8). Relabel its faces by that isomorphism, so that its induced map is exactly T. Lemma 3.4 then gives a realization of this representative, and Theorem 9.2 makes Φ_T satisfiable, contrary to hypothesis. Only the necessary direction of Theorem 6.3 is used; this corollary therefore does not depend on the abstract characterization in [5]. □

Remark 9.4 (No realizability conclusion)**.** Satisfiability of some Φ_T would *not* produce a polyhedron. Two gaps separate a model from a realization: the layers are necessary conditions and are not claimed to be sufficient, and even a complete uniform chirotope need not be realizable by real vectors — non-realizable uniform oriented matroids exist (see [18, Sections 2–3] for examples and discussion, and [7] for general realizability methods). This does not assert that any of our 59 formulas has a non-realizable model. A model would be a candidate to be examined, not a geometric counterexample.

9.1. **Case decomposition and cube trees.** Case decomposition permits parallel execution and bounded, resumable attempts. It is optional for correctness; the unsplit formula corresponds to depth $d = 0$. We first state the decomposition into the 2^d cases of one fixed depth d, and then its generalization to finite trees of cases, which is what the accompanying implementation records.

Proposition 9.5. *Let $p_1, \dots, p_d$ be distinct variables of Φ_T and, for $0 \le c < 2^d$, let U_c be the conjunction of the d unit clauses assigning p_m the m-th bit of c. If $\Phi_T \wedge U_c$ is unsatisfiable for every c, then Φ_T is unsatisfiable.*

Proof. A satisfying assignment of Φ_T satisfies U_c for exactly one c, namely the one read off its values at $p_1, \dots, p_d$. □

A fixed-depth partition can be refined locally: a case that exceeds the available time, proof size or checker memory can be split again while the other cases are left unchanged. Cube trees record precisely these successive refinements, so the implementation lets the tree grow where it is needed. For the rest of this subsection Φ is any CNF formula in the variables $1, \dots, V$, and an *assignment* is a map $\alpha\colon \{1, \dots, V\} \to \{0, 1\}$; it satisfies the literal x when $\alpha(x) = 1$ and the literal $\neg x$ when $\alpha(x) = 0$.

Definition 9.6 (Cubes and splits)**.** A *cube* is a set C of literals in which no variable occurs twice; $\varnothing$ is the *empty cube*. An assignment satisfies C when it satisfies every literal of C, and $\Phi \wedge C$ denotes Φ together with one unit clause for each literal of C. A *split* of a cube C is a sequence $W = (w_1, \dots, w_k)$, $k \ge 1$, of distinct variables none of which occurs in C. Its 2^k *children* are the cubes $C \cup \{\ell_1, \dots, \ell_k\}$ with $\ell_j \in \{w_j, \neg w_j\}$ for each j; each of them contains C and has $|C| + k$ literals.

Lemma 9.7 (A split partitions its cube)**.** *Let W be a split of a cube C. Every assignment that satisfies C satisfies exactly one child of C under W. Hence, if $\Phi \wedge C'$ is unsatisfiable for every child C', then $\Phi \wedge C$ is unsatisfiable.*

Proof. Let α satisfy C. Then α satisfies the child $C \cup \{\ell_1, \dots, \ell_k\}$ if and only if it satisfies each ℓ_j, and for each j it satisfies exactly one of w_j and $\neg w_j$. So it satisfies exactly one child: the one with $\ell_j = w_j$ when $\alpha(w_j) = 1$ and $\ell_j = \neg w_j$ when $\alpha(w_j) = 0$. If α satisfied $\Phi \wedge C$, it would therefore satisfy $\Phi \wedge C'$ for that child C'. □

Definition 9.8 (Cube tree)**.** A *cube tree* with root C_0 is a rooted tree whose nodes are labelled by cubes, such that the root is labelled C_0 and each node that is not a leaf, say labelled C, carries a split W of C and has exactly $2^{|W|}$ children, one for each child of C under W and labelled by it. It is *refuted* for Φ when $\Phi \wedge L$ is unsatisfiable for the label L of every leaf.

A cube tree is finite. Along every path down from the root the labels gain at least one literal at each step, and no cube has more than V literals, so the height is at most $V - |C_0|$; and each node has finitely many children.

Theorem 9.9 (Coverage)**.** *Let $\mathcal{T}$ be a cube tree with root C_0.*

(a) *Every assignment that satisfies C_0 satisfies the label of exactly one leaf of $\mathcal{T}$.*
(b) *If $\mathcal{T}$ is refuted for Φ, then $\Phi \wedge C_0$ is unsatisfiable. In particular, a cube tree with the empty root that is refuted for Φ shows that Φ is unsatisfiable.*

Proof. (a) By induction on the height of $\mathcal{T}$. If the root is a leaf, it is the only leaf and the claim holds. Otherwise let W be the split at the root, let $C_1, \dots, C_m$, $m = 2^{|W|}$, be the children of C_0 under W, and let $\mathcal{T}_t$ be the subtree below the child labelled C_t. Each $\mathcal{T}_t$ is a cube tree with root C_t and smaller height, and the leaves of $\mathcal{T}$ are those of $\mathcal{T}_1, \dots, \mathcal{T}_m$. Let α satisfy C_0. By Lemma 9.7, α satisfies exactly one C_t, and by induction it satisfies the label of exactly one leaf of $\mathcal{T}_t$. For $s \ne t$ every label in $\mathcal{T}_s$ contains C_s, because each label contains the label of its parent; as α does not satisfy C_s, it satisfies no leaf label of $\mathcal{T}_s$.

(b) If an assignment satisfied $\Phi \wedge C_0$, then by (a) it would satisfy the label L of some leaf, hence $\Phi \wedge L$, which is unsatisfiable by hypothesis. For the empty root, $\Phi \wedge \varnothing$ is Φ. □

Part (a) says that the leaves of a cube tree divide the region of its root into disjoint regions, with nothing left over; only part (b) is used below. Proposition 9.5 is the special

case of a tree with at most one split: for $d \geq 1$ its cases U_c are the children of the empty cube under the split $(p_1, \ldots, p_d)$, and for $d = 0$ the tree consists of its root.

The implementation does not store a tree but everything that was done: possibly several refutation attempts of one cube, and possibly several splits of one cube recorded by different machines. The following lemma shows that this is harmless.

Lemma 9.10 (Closed cubes)**.** *Let $\mathcal{R}$ be a set of cubes C for which $\Phi \wedge C$ is unsatisfiable, and let $\mathcal{S}$ be a set of pairs (C, W) of a cube C and a split W of C. Define, by induction on $V - |C|$, a cube C to be* closed *if $C \in \mathcal{R}$, or if there is $(C, W) \in \mathcal{S}$ such that every child of C under W is closed; the children have more literals than C, so this is already defined for them. Then every closed cube C is the root of a cube tree, refuted for Φ, whose leaf labels lie in $\mathcal{R}$ and whose splits come from $\mathcal{S}$. In particular $\Phi \wedge C$ is unsatisfiable for every closed cube C.*

Proof. By the same induction. If $C \in \mathcal{R}$, the tree with the single node C is refuted. Otherwise choose $(C, W) \in \mathcal{S}$ whose children are all closed, take for each child a refuted cube tree with that child as root, which exists by induction, and attach these trees below a root labelled C carrying the split W. The last claim is Theorem 9.9(b). □

In the implementation the role of $\mathcal{R}$ is played by the cubes with an admitted record of a checked refutation, so the lemma applies to the extent that these records are true (Remark 10.1), and that of $\mathcal{S}$ by the recorded splits (Section 10). To make its result reproducible it also selects one tree: at a closed cube it takes a refutation if there is one, and otherwise the first split in a fixed order whose children are all closed. Every such choice is a tree as in the proof, and which one is chosen does not affect the conclusion. A cube counts as closed only through a refutation of that cube or a split of it. The certifier does not infer closure of C from closure of a proper subset $D \subset C$ of its literals, although that inference would also be sound because $\Phi \wedge C$ implies $\Phi \wedge D$.

Corollary 9.11 (Conditional non-existence from cube trees)**.** *Suppose that for each of the* 59 *neighbourly maps T of Theorem 3.8 there is a cube tree with the empty root that is refuted for Φ_T. Then there is no neighbourly polyhedron with twelve faces satisfying* (H1)–(H3).

Proof. By Theorem 9.9(b) each Φ_T is unsatisfiable, and Corollary 9.3 applies. □

Remark 9.12 (A closed nonempty root settles only its region)**.** For a nonempty cube C_0, a refuted cube tree with root C_0 shows only that no satisfying assignment of Φ satisfies C_0: a statement about one region of the search space. Pilot computations are of this kind. A pilot may, for instance, take as its root one case U_c of a fixed decomposition, written as a cube, and refute a tree below it. That settles one of the 2^d cases and says nothing about the others; in particular it does not prove the formula. The implementation combines closed cubes into a refutation of Φ only as the leaves of a cube tree with the empty root — for example all 2^d cases of one decomposition — and then by Theorem 9.9 for that tree. Corollary 9.11 needs, for every one of the 59 formulas, a refuted tree whose root is the empty cube.

The choice of splitting variables affects cost only, never correctness. The default campaign first splits the empty cube of each formula on twelve order variables selected by an adaptation of the proof-prefix heuristic of [3]: bounded solver runs count occurrences of variables in added clauses. This gives $2^{12} = 4\,096$ initial cubes per formula. Subsequent splits use a bounded look-ahead heuristic, with four variables per split by default, or fewer when the candidate pool or probe budget requires it. The initial depth can be changed, including to zero; none of these policy choices is a hypothesis of the reduction. Because Lemma 9.7 and Theorem 9.9 hold for any choice, the heuristics need no correctness

argument of their own — but the splits must be recorded, since coverage is a statement about one tree.

10. The verification protocol

A statement of the form "for each of the 59 formulas, every leaf of a cube tree with the empty root is unsatisfiable" is only as good as the evidence that each leaf was decided, and decided about the right formula. We state the discipline the accompanying implementation enforces, since it is part of what would have to be reproduced.

The package `neighbourly` builds the formulas. `cubeproof` runs, records and certifies computations on fixed formulas; `faces12-campaign` binds these operations to the 59 maps and their pinned formula hashes. A campaign fixes the formulas, requested roots and tool identities. Workers can process disjoint shards, resume from their journals and transfer open cubes through recorded handoffs. Ownership determines who works on a region; only the admitted refutations and splits determine whether that region is closed. An unfinished or undelivered region remains open.

Three proof modes implement the same acceptance rule. In `transient` mode (the default), a proof is written, checked and deleted after its result is committed. In `retain` mode, accepted proofs are kept for later checking. In `stream` mode, a relay sends proof bytes directly to the checker and hashes the byte stream; it does not keep a proof file. If a resource limit prevents a check from completing, the leaf remains open and can be retried or subdivided. These execution choices do not change Φ_T or the coverage theorem.

Two programs, one of them trusted:

A CDCL solver (CaDiCaL) is used only as a *proposer*: for a leaf cube L it emits an LRAT refutation of $\Phi_T \wedge L$, which `cake_lpr` [22] checks, either from a proof file written once and sealed or streamed through a pipe. The correctness of `cake_lpr` is a machine-checked theorem carried down to machine code by the verified CakeML compiler. A leaf counts as refuted only when the checker's standard output is exactly the line `s VERIFIED UNSAT` followed by a newline, the process exits with code zero without a signal, and the CNF it read is the one regenerated from the formula and the cube. A leaf that unit propagation alone refutes gets a proof generated for it, which passes through the same checker. A solver's own claim of unsatisfiability is never accepted.

Identity of the object:

A formula is identified by the SHA-256 of its DIMACS text, and the bundle that carries it records the map, the layer string and the gauge face from which it was generated, so that it can be regenerated and compared byte for byte. A cube is written as its literals sorted by variable, so that its identity does not depend on the order in which it was reached. The CNF of a leaf is the DIMACS text of the formula, with the clause count in its header raised by the number of literals of the cube and one unit clause per literal of the cube appended in that order; every record names the hashes of the formula and of this CNF. Evidence about different formulas is never merged.

Attestation:

Each machine appends what it does to its own journal, a file of JSON records each of which carries its own hash and that of its predecessor: refutation attempts with their outcomes, splits, and satisfying assignments. A record of a refuted leaf is an *attestation*: it states that the checker accepted a proof with a given hash for a CNF with a given hash. By default the proof itself is deleted after the check. The hashes detect accidental damage and reordering inside a journal; they do not authenticate its writer, and they do not detect the replacement of a whole journal or the removal of its last records. The final head of each journal that a result relies on must therefore be published independently of the file.

Certification:
A certifier, which runs neither solver nor checker, reads the journals, recomputes the hash of every leaf CNF that a record names from the formula and the cube, admits only refutations checked by a checker binary that the campaign names by its hash, and evaluates closure as in Lemma 9.10. A formula counts as proved exactly when its empty cube is closed; a closed nonempty root is reported as such (Remark 9.12). A recorded satisfying assignment, which is checked against the clauses, is reported as an inconsistency if the empty cube of its formula is closed: by Lemma 9.10 and Theorem 9.9 both cannot be true, so one of the attestations or the implementation is at fault. The certificate names the selected tree's refutations, the cubes that remain open and one hash over the selected leaves. A second certifier, written independently from the specification, must produce the same result from the same journals. Agreement of two certifiers checks the evaluation of closure and the selection, not the honesty of the journals.

Independent verification:
The default mode does not retain proofs. Their total size depends on the eventual tree and solver runs; no storage estimate for a complete computation is claimed. An attested leaf is verified again in one of two ways. Where a proof was retained, the checker checks it again against the regenerated CNF, without the solver. Otherwise a second party recomputes the leaf: it regenerates the CNF and compares its hash with the record, runs the recorded solver again (or, for a leaf refuted by unit propagation, generates its proof again), has the checker verify the new proof, and compares the new proof's hash with the recorded one. Strict reproduction requires identical proof bytes under the recorded binary, options and execution profile; a difference fails that reproduction check. Determinism is tested on small workloads, not assumed as a mathematical theorem. A different checked proof still refutes the same leaf, but does not reproduce the recorded proof bytes. Either way the leaf is established anew, independently of the journal. Verifying a random sample of leaves tests the journals statistically and verifies only the leaves in the sample; verifying every selected leaf of every formula repeats the computation and verifies it completely. Neither a journal nor a proof is accepted without the corresponding check.

Exact controls and a reference encoder:
The identities of Section 5 are checked against exact rational arithmetic on explicit plane configurations: random ones in general position for Theorem 5.2, Lemma 5.5 and Lemma 5.7, and, for Layers B and I, which hold only for realizations, the tetrahedron and an exact rational realization of the Szilassi polyhedron [9] perturbed into general position, for which the predicates have separately computed truth values. Two reference checks serve different purposes. The package reference encoder shares the production encoder's conventions and compares clauses, detecting transcription errors. The additional reference in `tests/reference` is written from this article with its own bracket numbering and sign calculations. It reads the emitted clauses and the names of their variables, recognizes the auxiliary definitions by their truth tables, and compares the resulting Boolean polynomials over $\mathbb{F}_2$ with the predicates defined here. It also checks the substantive clauses, the gauge rank and reorientation invariance. This symbolic comparison covers all assignments of the bracket bits for each of the 59 fixed maps; separate rational tests check geometric interpretations on sampled arrangements and on the two known realizations, including relabellings and changes of gauge face. The reference reads the encoder's choices of link traversal and pinned corner and checks their admissibility; the pinned formula hashes fix the exact choices. Both references are maintained by this project. Neither constitutes external peer review or proves the geometric lemmas from agreement between programs.

The assurance argument has three distinct levels. The geometric reduction and the completeness of the case list are mathematical obligations. Correct encoding, formula identity and complete coverage are implementation obligations, supported here by exact controls and separate computations; coverage itself is Theorem 9.9, and the certifiers only evaluate its hypothesis on recorded evidence. Finally, the checker establishes unsatisfiability of the concrete CNF it reads, subject to the execution assumptions of its verification theorem; the small C layer through which the verified machine code reads its input files, and, when a proof is streamed, the pipe through which it arrives, lie outside that theorem; they are identified by the hashes recorded with each result. Formal verification of the checker does not discharge the first two levels. The solver need not be trusted to assert unsatisfiability: its proposed refutation is checked independently. The resulting theorem would nevertheless depend on the whole chain from geometry to the checked formulas.

Remark 10.1 (Attestation and verification are different guarantees)**.** The splits of a cube tree need no trust: whether a recorded split is a split in the sense of Definition 9.6 is decided by reading it, and its children are determined by it. Only the leaves carry computational content. A certificate that a formula is proved establishes the hypothesis of Theorem 9.9(b) for the selected tree at the level of attestation: each leaf is covered by a record of an accepted check. It does not show that the writer of an admitted record actually ran the checker on that CNF; a journal is not a proof. That a leaf is unsatisfiable is established independently of the journals only by checking a retained proof again or by recomputing the leaf. An independent verification of the whole computation therefore verifies every selected leaf of every one of the 59 formulas; a sample verifies only the leaves it selects.

11. Status, scope and what remains

11.1. **What has been done.** The geometric and encoding lemmas, the reduction and the coverage statements are proved here. The case-list theorem uses the published classification, whose concrete representatives are also recomputed and compared with the published data. The enumeration of Theorem 3.8 has been carried out and agrees in number with [1] and class by class with [14]. The encoder produces the 59 formulas of Definition 9.1, and the finite identities are tested by machine: Theorem 5.2, Lemma 5.5 and Lemma 5.7 against exact arithmetic on random rational configurations, Lemma 6.1 exhaustively on all tournaments with at most seven elements, which also reproduces the counts of Appendix A, Lemma 2.6 through the supplementary check of Remark 2.7, and the gauge of Corollary 8.2 by running all 2^{13} group elements and observing that they give the thirteen pinned brackets all 8 192 possible sign patterns within each orbit. The gauge selects one representative from each orbit; it does not identify the 8 192 patterns with different orbits. The tests of Lemmas 5.5 and 5.7 evaluate the encoder's own literals; the others check the identities independently of it. None of them replaces a proof given here.

11.2. **What has not been done.** A complete set of checked refutations has not been obtained: for none of the 59 formulas has a refuted cube tree with the empty root been recorded, let alone verified independently. Corollaries 9.3 and 9.11 therefore remain conditional: this paper does not settle the Question. Bounded pilot computations on restricted roots are not partial proofs of it beyond the regions they close (Remark 9.12). This version is a preprint of the reduction and of the protocol, released before the computation; the computation, and its independent verification, will be reported in a revised version. Clause counts and bounded tests do not provide a reliable estimate of the cost of all 59 instances. Cube trees support parallel and resumable attempts, but timeouts, resource limits and unrefuted leaves must remain explicitly open in any campaign report.

11.3. **Scope.** The reduction is stated for polyhedra satisfying (H1)–(H3). Dropping (H1) to allow two faces to meet in more than one edge or in an isolated additional point, dropping (H2) to allow faces with holes, or allowing self-intersections, changes the object: vertices of degree exceeding three become possible, the case list of Theorem 3.8 no longer applies, and the 59 formulas say nothing about those variants. Statements made here for general n, such as Proposition 2.5 and Lemma 3.6, are proved; the computations, and the tests of the implementation on maps with other numbers of faces, are finite checks of particular instances and establish nothing about polyhedra with other numbers of faces. We also note again the difference from the dual problem: Schewe's theorem [19] excludes polyhedral realizations of the 12-vertex genus-6 triangulations, which are combinatorially dual to our maps; realizability is not preserved by that duality, so it does not transfer a non-realizability result from one problem to the other.

Appendix A. Numerical data

A.1. **The case list.** The enumeration of Theorem 3.8 returns, for $n = 12$ with the link of one face fixed, 12 144 labelled orientable maps in 59 isomorphism classes. Their automorphism orders are

$\lvert\operatorname{Aut}\rvert$	1	2	3	4	6	12
classes	39	9	1	7	2	1

$$\sum_{C} \frac{264}{\lvert\operatorname{Aut}(C)\rvert} = 12\,144.$$

On the smaller sizes the same program returns one class for $n = 4$ ($\lvert\operatorname{Aut}\rvert = 24$), one for $n = 7$ ($\lvert\operatorname{Aut}\rvert = 42$, the Császár torus) and none for $n = 10$.

The automorphism orders are those stated in [8], Theorem 4. The 59 classes are those of the list of [14] for genus 6 and twelve vertices. The full correspondence, including all twelve-label bijections, is stored in `data/fixtures/lutz_correspondence.json`. The test applies them and compares the normalized file hash with the published file; the source URL, retrieval date and both raw and normalized hashes are recorded in `docs/SOURCES.md`. Keeping the correspondence with the data avoids a second, manually maintained numbering table in the article.

A.2. **Tournaments.** The counts behind Remark 6.2, obtained by exhaustive enumeration of all tournaments (the accompanying test suite recomputes them):

m	4	5	6	7
tournaments satisfying (GP)	48	384	3 840	46 080
transitive tournaments	24	120	720	5 040

matching $(m-1)!\,2^{m-1}$ and $m!$ respectively.

A.3. **The formula.** Sizes are as in Definition 9.1; the distribution of clause widths, counted from the emitted formula of the first map, is

width	1	2	3	4	total
S				23 760	23 760
H			15 840		15 840
A			2 880	4 320	7 200
B		528	2 112		2 640
I		324	1 056	1 344	2 724
gauge	13				13
Φ_T	13	852	21 888	29 424	52 177

The pattern behind the table: a three-term exclusive-or definition contributes eight clauses of width four, a two-term one contributes four clauses of width three, a conjunction contributes one clause of width three and two of width two, transitivity contributes clauses of width three, and the boundary clauses of B and the vertex clauses of I have width two. Exclusive-ors of more than three terms, which occur only in the orientation equation and in the four-term exclusive-or that defines each corner bit, are split into three-term ones with fresh variables; any assignment of the inputs extends uniquely to the fresh variables, so splitting changes neither the models nor the necessity argument. For the first map of the shipped list the DIMACS text has 5 257 variables and 52 177 clauses.

Appendix B. The accompanying software

The software is published in the repository

`https://github.com/pirozhkov-andrei/polyhedraproof`

This revision of the article was released with version 1.0.0rc1 of the software, under the tag `v1.0.0rc1`; the source archive of that release holds the article, the code, the data and the tests referred to here. The implementation is written in Python, needs nothing beyond the standard library at run time, and runs CaDiCaL and `cake_lpr` as external programs. Its code refers to the statements of this paper by their LaTeX labels — `lem:parity` for Lemma 5.10, for instance — which do not change when the numbering does. The correspondence itself, statement by statement, with the code that implements each one and the tests that check it, is the file `docs/THEORY_MAP.md` of the repository. It is generated from this paper's source and from a register of claims kept with the code, and a test fails when it is out of date or names a statement, a function or a test that does not exist; so it is kept there rather than copied here, where it could only go stale.

Use of generative artificial intelligence

Generative AI tools assisted with reviewing mathematical arguments, developing and checking the accompanying code, and drafting and revising the English exposition. These activities were part of the development process; AI-generated suggestions and reviews are not evidence of mathematical correctness, independent peer review, or formal verification. The author takes responsibility for the mathematical claims, the software, and the final text. The verification protocol described in Section 10 uses the published source code and the LRAT checker, not AI judgements.

References

[1] A. Altshuler, J. Bokowski and P. Schuchert, *Neighborly 2-manifolds with 12 vertices*, J. Combin. Theory Ser. A **75** (1996), no. 1, 148–162. doi:10.1006/jcta.1996.0069.

[2] E. Arseneva, L. Kleist, B. Klemz, M. Löffler, A. Schulz, B. Vogtenhuber and A. Wolff, *Adjacency graphs of polyhedral surfaces*, Discrete Comput. Geom. **71** (2024), no. 4, 1429–1455. doi:10.1007/s00454-023-00537-6; preliminary version in *Proc. 37th Int. Symp. on Computational Geometry (SoCG 2021)*, LIPIcs **189**, 11:1–11:17.

[3] Z. Battleman, J. E. Reeves and M. J. H. Heule, *Problem partitioning via proof prefixes*, in *Proc. 28th Int. Conf. on Theory and Applications of Satisfiability Testing (SAT 2025)*, LIPIcs **341**, 3:1–3:18. doi:10.4230/LIPIcs.SAT.2025.3.

[4] A. Biere, T. Faller, K. Fazekas, M. Fleury, N. Froleyks and F. Pollitt, *CaDiCaL 2.0*, in *Computer Aided Verification (CAV 2024)*, LNCS **14681**, Springer, 2024, 133–152. doi:10.1007/978-3-031-65627-9_7.

[5] A. Björner, M. Las Vergnas, B. Sturmfels, N. White and G. M. Ziegler, *Oriented Matroids*, 2nd ed., Encyclopedia of Mathematics and its Applications **46**, Cambridge University Press, 1999. doi:10.1017/CBO9780511586507.

[6] J. Bokowski and Kevin H., *Polyhedral embeddings of triangular regular maps of genus g, $2 \le g \le 14$, and neighborly spatial polyhedra*, Symmetry **17** (2025), no. 4, article 622. doi:10.3390/sym17040622.

[7] J. Bokowski and B. Sturmfels, *Computational Synthetic Geometry*, Lecture Notes in Mathematics **1355**, Springer, 1989. doi:10.1007/BFb0089253.

[8] M. N. Ellingham and C. Stephens, *Triangular embeddings of complete graphs (neighborly maps) with 12 and 13 vertices*, J. Combin. Des. **13** (2005), no. 5, 336–344. doi:10.1002/jcd.20048.

[9] B. Grünbaum and L. Szilassi, *Geometric realizations of special toroidal complexes*, Contrib. Discrete Math. **4** (2009), no. 1, 21–39. doi:10.55016/ojs/cdm.v4i1.61986.

[10] HackerPoet (CodeParade), *NeighborlyPolyhedra*, software repository, `https://github.com/HackerPoet/NeighborlyPolyhedra`. Accessed 15 October 2026. Related published account: [6].

[11] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Author's online edition: `https://pi.math.cornell.edu/~hatcher/AT/ATpage.html`.

[12] M. Jungerman and G. Ringel, *Minimal triangulations on orientable surfaces*, Acta Math. **145** (1980), no. 1, 121–154. doi:10.1007/BF02414187.

[13] F. Lazarus, *Linear Embeddings*, lecture notes, 2020, Section 2. `https://pagesperso.g-scop.grenoble-inp.fr/~lazarusf/Enseignement/effectiveGeom3.pdf`.

[14] F. H. Lutz, *The Manifold Page*, section *Surfaces*. `https://www3.math.tu-berlin.de/IfM/Nachrufe/Frank_Lutz/stellar/surfaces.html`. Retrieved 13 October 2026.

[15] G. H. Meisters, *Polygons have ears*, Amer. Math. Monthly **82** (1975), no. 6, 648–651. doi:10.1080/00029890.1975.11993898.

[16] G. Ringel, *Map Color Theorem*, Grundlehren der mathematischen Wissenschaften **209**, Springer, 1974. doi:10.1007/978-3-642-65759-7.

[17] G. Ringel and J. W. T. Youngs, *Solution of the Heawood map-coloring problem*, Proc. Nat. Acad. Sci. U.S.A. **60** (1968), no. 2, 438–445. doi:10.1073/pnas.60.2.438.

[18] M. Scheucher, *A SAT attack on Erdős–Szekeres numbers in $\mathbb{R}^d$ and the empty hexagon theorem*, Comput. Geom. Topol. **2** (2023), no. 1, 2:1–2:13. doi:10.57717/cgt.v2i1.12.

[19] L. Schewe, *Nonrealizable minimal vertex triangulations of surfaces: showing nonrealizability using oriented matroids and satisfiability solvers*, Discrete Comput. Geom. **43** (2010), no. 2, 289–302. doi:10.1007/s00454-009-9222-y; arXiv:0801.2582.

[20] T. Sulanke and F. H. Lutz, *Isomorphism-free lexicographic enumeration of triangulated surfaces and 3-manifolds*, European J. Combin. **30** (2009), 1965–1979. doi:10.1016/j.ejc.2008.12.016; arXiv:math/0610022.

[21] L. Szilassi, *Regular toroids*, Structural Topology **13** (1986), 69–80.

[22] Y. K. Tan, M. J. H. Heule and M. O. Myreen, *Verified propagation redundancy and compositional UNSAT checking in CakeML*, Int. J. Softw. Tools Technol. Transf. **25** (2023), no. 2, 167–184. doi:10.1007/s10009-022-00690-y; conference version: *cake_lpr: verified propagation redundancy checking in CakeML*, TACAS 2021, LNCS **12652**, 223–241.